You

Using these values, I want to perform linear classification, but I'm unsure which Machine learning model to use, can you help?

Selecting the input features and output labels

X = data[['GenHlth', 'MentHlth', 'PhysHlth']].values

y = data['Stroke'].values

Copilot

Sure! For linear classification, the following machine learning models are popular choices:

1. **Logistic Regression**: This is a simple yet effective linear model for binary classification. It's easy to interpret and can handle large datasets well.
2. **Linear Support Vector Machine (SVM)**: This model finds the hyperplane that best separates the classes in the feature space. It can be very effective, especially when there's a clear margin of separation between classes.
3. **Perceptron**: A type of linear classifier that updates its weights with each misclassification during training. It's a basic building block for more complex neural network models.

Here's a quick example of how you can implement logistic regression in Python using the scikit-learn library:

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

# Splitting the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Creating the logistic regression model
model = LogisticRegression()

# Training the model
model.fit(X_train, y_train)

# Making predictions
y_pred = model.predict(X_test)

# Evaluating the model
accuracy = accuracy_score(y_test, y_pred)
print(f'Accuracy: {accuracy}')
```

This example splits your data into training and testing sets, trains the logistic regression model, makes predictions, and evaluates the accuracy of the model. You can try different models and compare their performance to see which one works best for your data. Let me know if you need more information on any of these models or any additional help!
